PhD Project

CITY-SCALE MICROFINANCIAL FLOWS AND SOCIO-SPATIAL DISPARITIES IN SÃO PAULO

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1. Executive Summary

This project aims to develop an intra-urban diagnosis of economic centralities and socio-spatial disparities based on urban microfinancial flows observed through strictly aggregated electronic payment data within the São Paulo Metropolitan Region (RMSP). The objective is not to produce a commercial monitoring tool, but rather to test, through a reproducible and LGPD-compliant protocol, whether the spatio-temporal structure of these flows provides territorially meaningful information that complements conventional urban indicators such as accessibility, zoning, public facilities, and real-estate dynamics.

The project addresses three core challenges. First, it seeks to identify consumption centralities, territorial dependence gradients, and spatial discontinuities through robust indicators including spatial autocorrelation, clustering techniques, and polarization measures, while carefully accounting for aggregation bias (MAUP). Second, it aims to examine how these structures relate to inequalities in access to urban opportunities, particularly within peripheral and working-class areas, through triangulation with urban supply and transport networks. Third, it investigates the relationship between value circulation and the production of urban space by combining transaction-based flow signatures with real-estate proxies (ITBI) and planning layers to discuss mechanisms of territorial valorization and their socio-spatial implications.

Feasibility relies on a graduated access strategy (Plan A/B/C): aggregated Origin–Destination (OD) matrices under strong partnership conditions; fully sustainable place-of-expenditure analyses otherwise; and a contingency scenario mobilizing ITBI and urban open data should private datasets remain unavailable. Data governance is embedded within the research design through minimization principles, disclosure thresholds, top-coding procedures, joinability controls, and a strict separation between secure analytical environments and publishable outputs.

The expected contribution is twofold: (1) a spatially robust methodological framework for exploiting aggregated monetary flows without re-identification risks; and (2) a territorial interpretation of value circulation capable of informing ongoing debates at FAU-USP regarding inequality, urban governance, and the financialization and commodification of metropolitan space.

The analytical value of the project remains intact under Plan B and Plan C scenarios, as research questions primarily rely on aggregated territorial expenditure indicators and open urban datasets. Access to OD matrices enhances analytical depth but does not condition the production of scientifically publishable results nor the validity of the empirical design.

2. Problem Framing and Context

Payment infrastructures (cards, Pix, and related rails) generate massive transaction traces. However, the transformation of these flows into reproducible spatial knowledge regarding urban organization, centralities, gradients, discontinuities, and forms of territorial dependence, remains limited and is rarely governed for simultaneous scientific, public, and institutional use. The challenge does not lie in data availability but in their epistemic status: they are often mobilized ad hoc for operational purposes without transparent spatial robustness protocols, systematic articulation with urban layers (zoning, built environment, accessibility), or clearly defined compliance frameworks.

This issue is particularly acute in the RMSP, where socio-spatial inequalities shape access to urban opportunities and where dynamics of valorization and devalorization are intertwined with public policy, peripheral urbanization, infrastructure development, and investment cycles. The objective is therefore not to map expenditures per se, but to describe where consumption flows concentrate, which territorial asymmetries become visible (dependence on a limited number of poles, localized fragilities), and how these structures interact with established urban diagnostics.

Beyond conventional inequality analyses, the RMSP exhibits patterns of functional dependence between peripheral territories and economic centralities, unequal distribution of urban resources, and only partial visibility of everyday economic practices within traditional statistical systems. In this context, payment infrastructures can be conceptualized as an informational layer revealing the ordinary structuring of the metropolis: differential territorial attractiveness, access constraints to services, and polarization of consumption circuits. The research therefore seeks to determine whether everyday monetary circulation renders observable urban dynamics that have been studied through other indicators but rarely captured at this functional level of granularity.

The scientific problem lies at the intersection of three dimensions. First, characterizing centralities and disparities from aggregated transactions while distinguishing robust spatial structures from measurement artefacts such as cash-versus-electronic bias and uneven coverage. Second, producing indicators and segmentations resilient to spatial analysis pitfalls, particularly MAUP, and stable over time through out-of-sample and spatio-temporal validation. Third, situating findings within a critical urban political economy perspective that conceptualizes payment systems as infrastructures capturing and reconfiguring urban value.

The articulation with urban space production and real estate dynamics is central. By combining transaction aggregates with urban layers and real-estate proxies (ITBI), the project tests plausible mechanisms of co-evolution between economic centrality, monetary circulation, and valorization trajectories without shifting toward market prediction. Finally, the problem framing acknowledges a practical constraint: scientific value depends on data governance. A graduated

access strategy and privacy-by-design framework constitute preconditions for producing publicly contestable knowledge.

Within this perspective, observed economic centralities are not merely functional configurations but potential components of urban space production interacting with infrastructures, public policies, and investment trajectories. The analysis therefore situates monetary flows within a broader urban chain linking value circulation, material transformation, and sectoral territorial restructuring.

The following figure illustrates the type of analytical visualization targeted by the project, a reconstruction of intra-urban consumption flows from aggregated transaction data, enabling the visualization of commercial centralities, territorial dependencies, and socio-spatial configurations. The project proposes to develop this empirical approach within the São Paulo metropolitan context.

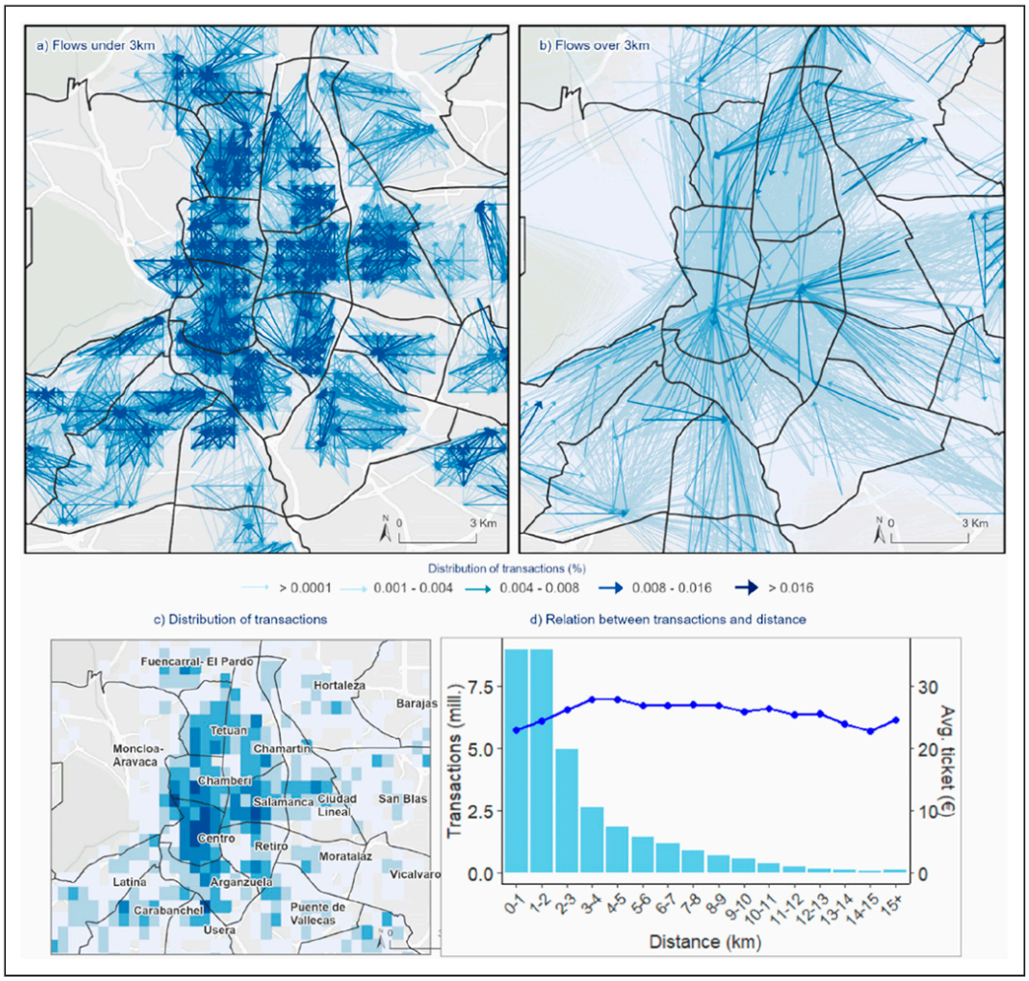


Figure 1. Origin–Destination Matrix. (a) Flows between nodes >3 km apart; (b) Flows between nodes >3 km; (c) spatial distribution of transactions; (d) relationship between average ticket (points) and number of transactions (bars) with distance.

Source : de las Obras-Loscertales, J., García-Palomares, J. C., Ramiro-Fariñas, D., Hernández Martín-Caro, M. S., & Gutiérrez, J. (2026). Intra-urban consumption flows and sociospatial disparities: An analysis based on bank card transaction data. EPB: Urban Analytics and City Science, 0 (0), 1–19. https://doi.org/10.1177/23998083261420309

3. Research Question, Sub-questions and Hypotheses

3.1 Central Research Question

How is intra-urban monetary value circulation, observed through aggregated electronic payment indicators, spatially structured within the São Paulo Metropolitan Region, and how do these structures relate to socio-spatial inequalities and to processes of urban space production and valorization, particularly real-estate dynamics, within an LGPD-compliant analytical framework?

3.2 Sub-questions

- Q1 Centralities and urban economy: Which economic centralities and territorial dependence gradients emerge from payment flow structures (densities, temporal patterns, attractors), and to what extent do these configurations coincide with or diverge from "official" centralities derived from employment distribution, transport networks, zoning, and public facilities?

- Q2 Inequalities and access: To what extent do monetary flow structures reveal asymmetries in access to urban opportunities (services, commerce, payment infrastructures) and territorial vulnerabilities, particularly in peripheral and working-class areas?

- Q3 Urban space production and financialization: How do dynamics of monetary circulation and economic centrality relate to trajectories of valorization and devalorization and to real-estate production (proxied through ITBI), and which urban or policy mechanisms plausibly explain these co-variations (infrastructure investment, regulation, urban projects, investment cycles)?

3.3 Hypothesis

Hypothesis	**Summary Formulations**
H1 (Spatial structure)	Electronic payment activity exhibits stable spatial autocorrelation (Moran's I / LISA) and a hierarchy of centralities not reducible to transport or employment networks alone, thereby revealing consumption-specific centralities.
H2 (Inequalities)	Under comparable urban characteristics, working-class peripheral areas display distinctive flow signatures, limited commercial diversity, dependence on a small number of poles, and temporal volatility, associated with constrained access to payment infrastructures and territorial opportunities.
H3 (Real estate and urban production)	Areas experiencing intensification and diversification of monetary flows precede or accompany signals of real-estate

	valorization (ITBI transactions), suggesting heterogeneous co-evolution between economic centrality, financial circuits, and urban space production.

4. Theoretical Positioning

The project is positioned at the intersection of (i) urban geography and urban economics (centralities, accessibility, metropolitan planning and governance), (ii) critical urban political economy (value circulation, financialization and commodification, inequality), and (iii) quantitative geography and spatial methods (spatial autocorrelation, spatial econometric modeling, MAUP robustness). Its originality lies in combining sensitive transactional material with a reproducible analytical protocol and a privacy-by-design governance framework to generate results that are simultaneously scientifically contestable and politically interpretable.

5. Data, Access Strategy and Sources

The research relies on a graduated Plan A/B/C data access strategy designed to avoid dependency on fine-grained data promises.

Plan A (ideal, strong partnership): Aggregated transactions by spatial unit and time interval, with aggregated activity categories and Origin–Destination matrices exclusively available in anonymized and aggregated zonal form.

Plan B (likely, partial partnership): Absence of OD matrices; implementation of place-of-expenditure analyses (intensity, diversity, volatility, polarization, spatial autocorrelation) triangulated with urban layers and ITBI datasets.

Plan C (fallback, fully feasible): ITBI combined with urban open data and public payment statistics, enabling a refocusing on urban production, valorization processes, and financial infrastructures through spatially robust methods.

Core data sources include ITBI datasets from the Municipality of São Paulo, GeoSampa urban layers, SPTrans GTFS transport data, IBGE socio-demographic indicators, and Banco Central do Brasil Pix statistics.

5.1 Dataset Inventory (open-data backbone and partnership targets)

Base	Contenu	Accès (cible)	Source

Pix payments	Aggregated transaction volumes, values, and user statistics	BCB / sandbox / open stats	Banco Central do Brasil (Pix)
Card payments	Aggregated flows by place, time, and MCC category	Commercial banks	Partnerships with Commercial banks or payment networks
ITBI (real estate transactions)	Transaction value, location, and date	Open data municipal	Prefeitura de São Paulo (Fazenda/GeoSampa)
Socio-economic indicators	Censo IBGE + local indicators	Open data / institution	IBGE / Prefeitura / USP

6. Data Governance and LGPD Compliance

The research design establishes a strict separation between secure analytical environments and publishable outputs. Core governance principles include minimization (no individual-level data), disclosure thresholds and suppression of rare cells, adaptive spatial-temporal aggregation where volumes are limited, top-coding of extreme values, and joinability controls preventing reconstruction of rare sub-groups through orthogonal data cuts. All exports undergo compliance review procedures and are documented through pipeline versioning and comprehensive metadata.

7. Methodology and Empirical Design

Spatial units: A dual-unit approach combining fine administrative divisions and multi-resolution regular grids to assess result stability under MAUP conditions.

Indicator architecture: Four analytical blocks covering local intensity, centralities and polarization, inequalities of access, and production and valorization dynamics.

Analytical outputs include (i) multi-scale mapping of metropolitan consumption centralities; (ii) identification of territorial dependence gradients and peripheral configurations; (iii) typology of territorial regimes combining economic centrality, accessibility, and transaction signatures; and (iv) cross-analysis with real-estate dynamics enabling discussion of co-evolution mechanisms between monetary circulation and urban transformation.

Validation strategies include out-of-sample temporal validation, spatial block cross-validation to prevent leakage effects, sensitivity tests across scales and aggregation rules, and retention of only stable analytical patterns

7.1 Concepts and measures (operational definitions)

Concept	Operational definition	Measure / proxy
Centralities	Concentration of value and economic attractiveness	Transaction density, OD network centrality
Corridors	Value circulation axes between residential and commercial zones	OD matrices, inbound and outbound flows
Access inequalities	Asymmetries in access to services and centralities	Travel-time distance versus flows, spatial heterogeneity of flows
Financialization	Transformation of spaces through financial valorization logics	Composite index (average ticket size, growth, concentration, real-estate signals)
Resilience	Capacity to absorb shocks and sustain local activity	Stability (CV), seasonality, diversity (entropy)

7.2 Validation and robustness plan (excerpt)

Hypothesis	Main test	Robustness check	Expected result
H1 Spatial non-randomness	Moran I / LISA	Monte Carlo permutation	Significant clustering
H2 Hierarchies	OD centrality / distribution	Alternative W matrices	Stable hierarchical structure
H3 Signatures	Clustering + PCA	Sensitivity to MCC aggregation	Interpretable clusters
H4 Temporal precedence	Dynamic panels (lag t-1)	3 versus 4-year windows	Flows precede real-estate signals

8. Expected contributions

Methodological contribution: a reproducible protocol to exploit aggregated monetary flows while controlling for MAUP, spatial dependence, and joinability risks.

Empirical contribution: a robust mapping of consumption centralities and territorial dependence gradients within the RMSP, enabling the identification of access asymmetries and local vulnerabilities.

Theoretical contribution: a situated discussion of the relationships between value circulation, payment infrastructure governance, and the production and valorization of space (ITBI), in dialogue with debates on inequality and financialization.

Public contribution: more broadly, the project engages with debates within FAU USP concerning socio-spatial inequalities, metropolitan restructuring, and urban governance by introducing an empirical perspective centered on everyday monetary circulation as a constitutive dimension of contemporary urban organization.

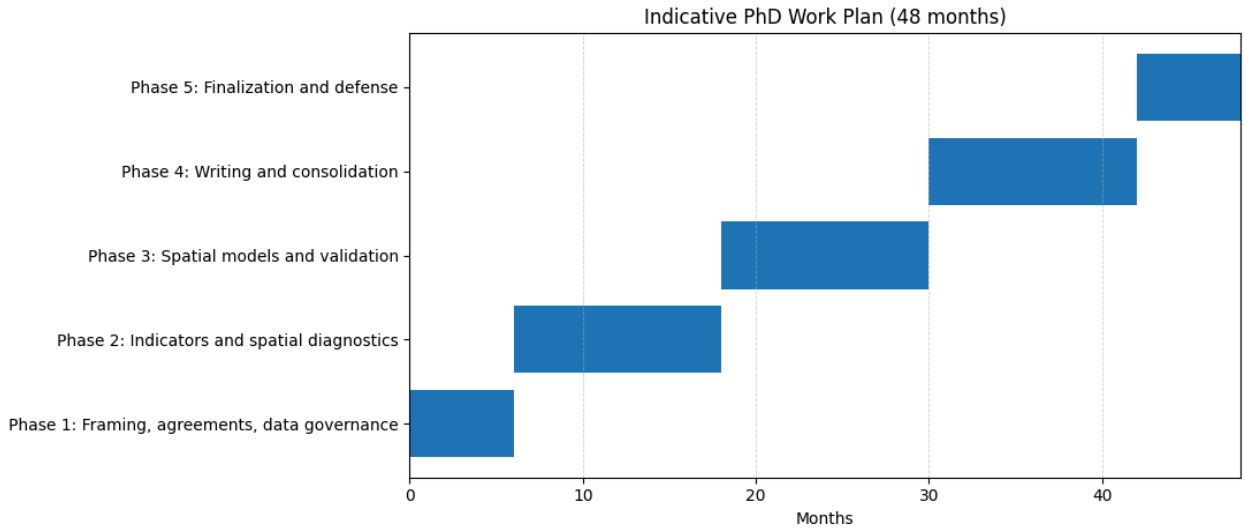
8.1 Alignment with FAU USP research axes (three entry points, one project)

- Entry point 1 Right to the city / governance / inequalities: reading monetary flows as indicators of territorial constraints, dependence on centralities, and access vulnerabilities without individual targeting, while discussing the effects of urban policies and metropolitan restructuring processes.

- Entry point 2 Planning and urban economics: construction of a reproducible empirical diagnosis of consumption centralities and accessibility structures using spatial models and strong validation procedures including MAUP checks, spatial dependence testing, and spatio-temporal blocking.

- Entry point 3 Real-estate production and valorization cycles: articulation of monetary flows, real-estate proxies (ITBI), and planning layers to test mechanisms of co-evolution between economic centrality, valorization processes, and sectoral transformations within a critical perspective rather than market prediction.

9. Work plan, risks, and feasibility



9.1 Risk register (excerpt)

Risk	Level	Mitigation	Plan B
Delayed data access	High	Multi-partner strategy (bank, network, BCB)	Aggregated-only scenario
LGPD constraints	Medium	Audit, aggregation, suppression of rare categories	Reduced granularity
Coverage bias	Medium	Socio-economic calibration and sensitivity testing	Multi-source comparisons
Computational complexity	Medium	Cloud computing and ETL optimization	Temporal subsampling

Appendix :

A. Glossary

PIX
Brazilian instant payment system operated and coordinated by the Central Bank of Brazil, enabling real-time transfers between individuals, firms, and institutions.

MCC
Merchant Category Code, a standardized classification of merchants by activity category used to group and analyze transactions by economic sector.

OD
Origin Destination matrix describing aggregated and anonymized flows linking an origin zone to a destination zone (for example residence to place of expenditure).

Moran's I / LISA
Global and local spatial autocorrelation tests used to identify territorial clusters of similar or contrasting values.

MAUP
Modifiable Areal Unit Problem referring to biases generated by the choice of spatial aggregation scale or unit shape in territorial analysis.

LGPD
Brazilian General Data Protection Law (Lei 13.709/2018) regulating the collection, processing, and dissemination of data potentially enabling individual identification.

ITBI
Municipal tax applied to real-estate transactions in Brazil used as an empirical proxy for market dynamics and valorization processes.

Privacy by design
Approach integrating data protection requirements from the earliest stages of research protocol design including aggregation rules, publication thresholds, recomposition controls, and separation between analysis and publication environments.

Consumption centrality
Capacity of a territory to attract expenditure flows and everyday economic activities, revealing its functional role within metropolitan organization.

Urban microfinancial flows
High-frequency, low-value monetary flows used as indirect indicators of ordinary economic activity and territorial dynamics.

Top coding
Technique consisting of capping extreme values to reduce identification risks and stabilize statistical distributions.

Spatio-temporal validation
Robustness procedure testing results across time periods and territories distinct from those used for estimation.

Territorial regimes
Spatial configurations characterized by specific combinations of centrality, dependence, accessibility, and valorization used as an interpretative analytical framework.

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